

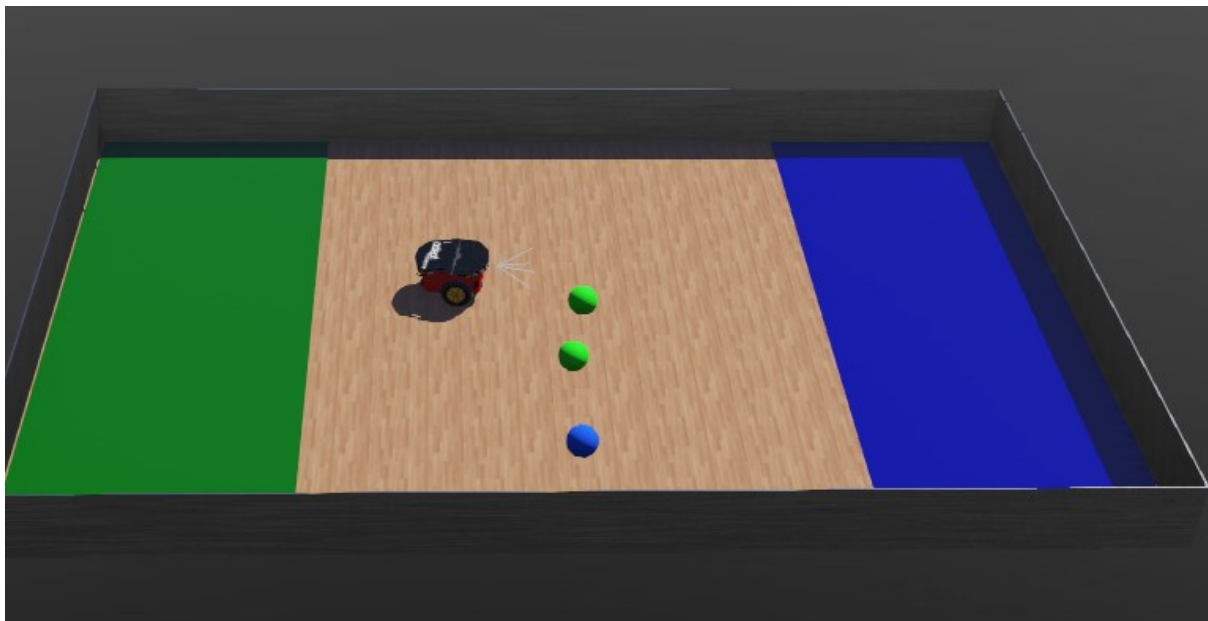
## CSCI291 Project

### PROJECT DESCRIPTION

This project is an **open-ended** mobile robot application using the **Webots** simulation platform. It aims to design and implement, in **C language**, a fully autonomous mobile robot. Using its **embedded camera**, the robot should detect balls of different colors, recognize their color, and push them to the corresponding matching area.

You will be provided with the Webots project folder, which includes the robot model ([Pioneer-3dx](#)) and its working environment containing several colored balls and a matching set of color-coded goal sections on the side walls (see Figure 1). Your task is to write the controller for the robot to operate as a smart sorter, intelligently locating and pushing each ball to its corresponding goal section. The core challenge lies in building a robust sequential logic that ensures all balls are processed without collisions. The robot must navigate from one ball to the next and then to the correct goal, all while avoiding contact with other balls. It is assumed that the color of each side remains fixed.

Upon successfully sorting every ball, the robot must provide a final tally of the number of balls processed, demonstrating both a complete understanding of its environment and successful task completion.



**Figure 1:** Webot Robot Environment.

### PROJECT REQUIREMENTS

The following are the formal requirements for the successful completion of this project. The final C program must satisfy all of these conditions.

1. **Controller Framework:** The entire robot's logic must be contained within a single C program that operates as a Webots controller for the Pioneer 3DX robot.
2. **Random Color Selection:** At the beginning of each simulation run, the program shall use a C random color generator to select one of the two ball colors as its first target\_color. The selected color must be output to the console for verification.

3. **Vision Processing:** The controller must process image data from the robot's camera to identify the color and pixel location of all four balls present in the arena.
4. **State-Based Behavior:** The robot's actions shall be governed by a **finite state machine**, with clear transitions between states such as `SEARCHING_FOR_BALL`, `MOVING_TO_TARGET`, `AVOIDING_NON_TARGET`, and `PUSHING_TOWARDS_GOAL`.
5. **Targeted Navigation:** The robot must autonomously navigate to the single, correctly identified target ball using its vision data.
6. **Obstacle Avoidance:** The robot should not hit the wall, or a ball of other color. The robot can enter the colored side.
7. **Goal Completion:** The robot must successfully push the target ball into the designated 1.5-meter goal section, at which point the mission is considered complete.
8. **Robustness:** The controller must be capable of successfully completing the task across multiple simulation runs with varying starting positions for both the balls and the robot, as well as different randomly selected first target colors.

## CONSTRAINTS

Please follow the constraints below:

- Do not modify the environment or the robot.
- Do not hard code any motions.
- Do not use any API.

## SUBMISSION

For the submission please include:

1. An up to 5 to 10-minute narrated video in which you demonstrate the operation of the robot (this section may be recorded at a high speed), explain the algorithms you have used with reference to your code, and highlight any portions of the code that were AI-generated or taken from a third party. Please upload the video as “unlisted” on youtube or One-drive and only submit the link.
2. Up to a 4-page report explaining the ball color identification, ball location identification, obstacle avoidance techniques, motion control and planning. You may also include any other methods you have used. The report should conclude with evaluation and reflection sections to assess the project’s achievements and reflect on your project development, highlighting the main lessons learned from the design and implementation phases.
3. Full C Code duly commented and the Webots project folder.

**MARKING SCHEME**

| <b>Marking item</b>              | <b>Mark</b> |
|----------------------------------|-------------|
| Robot movement (collision free)  | 6           |
| Sensor readings, camera setup    | 4           |
| Color and ball identification    | 7           |
| Pushing a ball into colored zone | 10          |
| Progress                         | 5           |
| Report                           | 3           |
| Video Demo                       | 5           |
| <b>Total</b>                     | <b>40</b>   |

Please note that generated work will get a penalty.